

DANIEL ZHANG

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EDUCATION

Bachelor of Engineering, Mechatronics Engineering CO-OP

McMaster University, Lvl III

Hamilton, ON

Expected Apr 2028

- Notable Coursework: Analog & Digital Circuits, Statics, Dynamics, Electricity and Magnetism, Thermodynamics, Embedded Systems, Engineering Mathematics, Data Structures & Algorithms

SKILLS

Industry & Safety: Project Engineering, Process Flow Automation, Technical Writing, WHMIS, Lockout/Tagout

Design, Modeling & Hardware: CAD (SOLIDWORKS — CSWA Certified, Inventor, Fusion 360), NI Multisim, Embedded Systems, STM32, Soldering, 3D Printing, GD&T

Software: Python, C/C++, MATLAB/Simulink, Julia, SMath, Maple, FlexPDE (FEA), Microsoft 365, Power Automate

EXPERIENCE & INVOLVEMENT

Mechanical Sub-Team Member — Wheel Assembly

McMaster Chem-E-Car

Hamilton, ON

Sep 2026 - Present

- Engineering a custom wheel assembly in SolidWorks for a $40 \times 30 \times 18$ cm chemically powered vehicle designed to carry an up to 500 mL water payload across a 33 m track.
- Collaborating with a team of 3 to optimize wheel assembly layout, ensuring compliance with AIChE size constraints and payload requirements for competition readiness.
- Engineered an independent double wishbone suspension to isolate dynamic weight transfer, stabilizing the vehicle chassis to prevent false triggers on optical stopping sensors.

Nuclear Engineering Co-op Student

Eclipse Automation

Cambridge, ON

Jan 2026 - Present

- Drafted the development of high-consequence engineering deliverables (FMEAs, FAT Plans, System Design Descriptions) for CSA N299.2 and N299.3 rated tools, driving successful milestones for the Pickering, Cernavodă, and Qinshan nuclear reactor refurbishment projects.
- Performed Allowable Strength Design (ASD) calculations for specialized nuclear tools for the Pickering (RFBR) refurbishment project, determining required safety margins to prove the theoretical viability and operational safety of mechanical systems.
- Delivered project engineering support for Design Verification Testing (DVT) of 4 specialized manufacturing tools, partnering with toolmakers, automation technicians, electricians, and clients to validate system performance against design requirements and facilitate customer sign-off.
- Facilitated CNSC regulatory approval for CNL's \$100M SBWRS project by verifying 5,000+ technical requirements, and engineered an automated workflow to audit a library of 1,000+ engineering standards and flag discrepancies.

Tutor

Academy for Mathematics & English

Richmond Hill, ON

Jul 2024 - Dec 2025

- Provided one-on-one tutoring in mathematics and physics for elementary, high school and university-prep students, simplifying complex technical concepts into clear, practical explanations.
- Designed personalized lesson plans and study materials to target individual learning gaps, improving students' test scores and conceptual understanding over multiple sessions.

PROJECTS

12-DOF Quadruped Robot

Aug 2026 - Present

- Designing a 12-DOF quadruped in Autodesk Fusion, integrating a 3D-printed chassis, MG90S servos, a PCA9685 driver, and a Raspberry Pi Zero 2 W.
- Developing a custom IK solver for stable omnidirectional gait patterns, with OpenCV camera integration for autonomous obstacle avoidance.

Computer Vision-Controlled Robotic Arm

Apr 2025

- Prototyped a 2-DOF robotic arm with a team of 3 in a 12-hour hackathon, pairing a Python OpenCV vision pipeline with an ESP32 to track hand movements in real time.
- Engineered embedded C++ firmware to process coordinate packets over UDP and generate step pulses, tuning motion profiles to eliminate jitter and ensure smooth multi-axis tracking.